NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
NASA SHARED SERVICES CENTER

RECOMMENDATION AND DETERMINATION TO SOLICIT FROM ONE SOURCE
OR BRAND NAME

PICK ONE

Sole Source:
☐I recommend that NASA, NASA Shared Services Center negotiate with Vendor Name only for Enter Supply or Service to be procured. The total estimated cost of this effort is Enter Dollar amount and the estimated period of performance or lead-time for delivery is Date.

OR

Brand Name Justification:
☒I recommend that NASA, NASA Shared Services Center negotiate with only with vendors who can provide the subject brand name from Stephens Analytical for MicroView Mini 6000 Hygrometers. The total estimated cost of this effort is [redacted] and the estimated period of performance or lead-time for delivery is 9/11/2026.

This recommendation is for the acquisition of supplies or services determined to be reasonably available from only one source/only one manufacture. Competition is impractical for the following reasons (please answer the questions below):

1. **What are the reasons for soliciting only the suggested source/suggested brand name?** Please provide the unique characteristics of the items or services along with the supporting rationale and the market research performed to substantiate your reason for the suggested source/brand name.

   This product line from Stephens Analytical has a unique combination of accuracy, response time, and ease of use for the price point. Fast-response, high accuracy systems exist in the form of flight-rated spectrometers, however, they cost 5x the cost of the proposed hygrometer. FLYT WVSS are spectrometer units which are used at PSL, however, they cost $70-80K per unit. Simpler units may be cheaper and easier to use, but their accuracy would result in dewpoint temperature measurement errors in the 10x10 SWT which directly correlates to increased Mach number uncertainty. Vaisala was considered for the primary unit, however, the accuracies of their sensors would correspond to higher than desirable Mach number uncertainty contributions in the facility. Chilled-mirror units are temperamental and were removed from the facility due to maintainability in the past. Edge-Tech sells chilled-mirror units which are challenging to maintain and use outside of well-controlled laboratory environments. The Push-Purge system from Stephens Analytical allows for efficient verification of the humidity levels being sample, and the accuracy across the operating range desired is NIST-traceable and on-par with chilled-mirror units.

2. **What is the impact to the government if another vendor/brand is used?** Include any dollar, schedule, or compatibility impacts if possible.

   If another vendor and/or brand name of hygrometer could result in excessive test time required to achieve a sufficiently stable dewpoint measurements or excessive error in dewpoint temperature measured, thus resulting in Mach number errors for the facility and customers. If a chilled mirror system is chosen, downtime for sensor maintenance could add to the facility workload and downtime.

ADDITIONAL COMMENTS:

The units of interest from Stephens Analytical are capable of measuring humidity levels of 5-6000 ppm(V) with NIST-traceable calibration, 4-20 mA output, and <1 min response time through a remote-activated Push Purge feature which is proprietary to the company.

[redacted] [redacted]

[redacted]